

A Multi-Sensor IoT Framework for Real-Time Elderly Health Monitoring with Adaptive Threshold-Based Alert Reduction

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Abstract—As the elderly population continues to grow worldwide, there is an increasing need for accurate, affordable, and real-time health monitoring systems that support independent living. This paper presents an Internet of Things (IoT)-based elderly health monitoring system that continuously monitors vital health parameters such as heart rate, blood oxygen saturation (SpO_2), body temperature, and fall detection. The proposed system is built around the ESP32 microcontroller, which collects physiological data from the MAX30102 pulse oximeter sensor, body temperature data from the MLX90614 infrared temperature sensor (or DS18B20 if that is what you finally use), and movement data from the MPU6050 accelerometer and gyroscope for fall detection. The measured values are displayed locally on a 0.96-inch OLED display and an active buzzer provides an immediate alert whenever abnormal health conditions or falls are detected. The collected data is transmitted over Wi-Fi to a cloud server, where it is visualized through a real-time monitoring dashboard. In emergency situations, the system can also notify caregivers through the dashboard or, when equipped with a SIM800L GSM module, by sending SMS alerts. The proposed solution is low-cost, energy-efficient, and easy to deploy, making it suitable for home-based elderly care, remote patient monitoring, and healthcare applications in resource-constrained environments.

Index Terms—Internet of Things, elderly health monitoring, fall detection, ESP32, MAX30102, MPU6050, MQTT, adaptive thresholding, SpO_2 , wearable sensors

I. INTRODUCTION

The rapid growth of the elderly population has created a significant demand for intelligent healthcare systems that

enable continuous monitoring and timely medical assistance. According to the United Nations, the global population aged 65 years and above is expected to increase substantially over the coming decades, leading to greater healthcare challenges and increasing pressure on existing medical services [1]. Many elderly individuals, particularly those living alone, are vulnerable to health emergencies such as abnormal heart rate, low blood oxygen levels, sudden changes in body temperature, and accidental falls [2], [3]. Delayed medical intervention in such situations can result in serious health complications and, in severe cases, loss of life.

Traditional healthcare systems primarily rely on periodic medical examinations and manual reporting of symptoms, which are often insufficient for providing continuous monitoring and immediate emergency response. Although several commercial health monitoring devices are available, many are expensive, consume considerable power, or require continuous user interaction, making them less suitable for long-term monitoring of elderly individuals. Furthermore, many existing systems focus on monitoring a single health parameter, limiting their ability to provide a comprehensive assessment of a patient's condition.

Recent advancements in the Internet of Things (IoT), embedded systems, and wireless communication technologies have enabled the development of affordable and efficient remote healthcare solutions. Low-cost microcontrollers such as the ESP32, combined with biomedical sensors and cloud-

based platforms, allow continuous acquisition, processing, and transmission of physiological data in real time. These technologies facilitate remote monitoring by caregivers and healthcare professionals while enabling early detection of abnormal health conditions.

This paper proposes an IoT-based Elderly Health Monitoring System that continuously monitors heart rate, blood oxygen saturation (SpO_2), body temperature, and fall detection using the MAX30102, MLX90614, and MPU6050 sensors integrated with the ESP32 microcontroller. The monitored data are displayed on an OLED display and transmitted to a cloud-based dashboard through Wi-Fi for real-time remote monitoring. In emergency situations, the system generates local alerts using a buzzer and can send SMS notifications to caregivers through the SIM800L GSM module. The proposed system aims to provide a low-cost, portable, energy-efficient, and reliable solution for continuous elderly health monitoring, improving patient safety and enabling timely medical intervention.

Contributions

The main contributions of this work are as follows:

- Development of a low-cost IoT-based elderly health monitoring system capable of continuously monitoring heart rate, blood oxygen saturation (SpO_2), body temperature, and fall detection using the ESP32 microcontroller.
- Integration of multiple biomedical sensors, including the MAX30102 pulse oximeter, MLX90614 infrared temperature sensor, and MPU6050 accelerometer and gyroscope, to provide comprehensive real-time health monitoring.
- Implementation of a cloud-based remote monitoring platform that enables caregivers to access real-time health data through a web dashboard using Wi-Fi connectivity.
- Integration of an emergency alert mechanism using an active buzzer and an optional SIM800L GSM module to notify caregivers immediately when abnormal health conditions or falls are detected.
- Design of a portable, energy-efficient, and cost-effective healthcare monitoring solution suitable for home-based elderly care and remote patient monitoring.

The rest of the paper is organised as follows. Related work is discussed in Section II. Section III presents the architecture and hardware design. Section IV explains the proposed AMPV algorithm. Section V provides the experimental results. Section VI discusses limitations and future work. The conclusion is drawn in Section VII.

II. RELATED WORK

Research on IoT-based health monitoring has become very important because more and more people need to have their health checked all the time from far away and need help quickly if there is an emergency. Recent studies have looked at combining biomedical sensors, wireless communication technologies, cloud-based monitoring platforms and small computer systems to keep track of health signs like heart rate, blood oxygen levels (SpO_2) body temperature and when

someone falls. This section looks at the important research about IoT-enabled healthcare for older people, wearable sensing technologies, remote patient monitoring and ways to send alerts when there is a problem. It shows what is good, about these systems what is not so good and why the new system is needed.

A. Fall Detection

Mubashir et al. [4] conducted a comprehensive survey of fall-detection techniques and categorised approaches as wearable, ambient (camera- or microphone-based), or hybrid. Accelerometer-only systems achieved sensitivities in the range of 80–92% but suffered from high false-alarm rates when subjects performed rapid daily movements such as sitting down forcefully or stumbling without falling. Incorporating a gyroscope alongside the accelerometer reduced false positives [18], yet the authors noted that single-modality wearables remain insufficient for real-world deployment without activity-context awareness.

B. Wearable Vital-Sign Monitoring

Patel and other researchers [5] looked at sensors that are used to keep track of health all the time. They found that heart rate and SpO_2 are the useful signs for checking health at home. They said that photoplethysmography sensors that go on the wrist or finger give a mix of being accurate and easy to use.. They added that motion problems in the data need to be fixed during processing. Later studies, in the IoT area have used the MAX30100 and MAX30102 sensors for checking pulse and oxygen levels on the wrist in devices.

C. IoT-Enabled Healthcare Platforms

The emergence of low-power Wi-Fi System-on-Chip (SoC) devices, such as the ESP8266 and ESP32, has significantly accelerated the development of IoT-based remote healthcare systems. These microcontrollers, combined with cloud platforms such as ThingSpeak, Adafruit IO, Firebase, and AWS IoT, enable real-time acquisition, transmission, and visualization of physiological data. Such systems allow caregivers and healthcare professionals to monitor patients remotely and receive timely notifications during emergency situations. Despite these advancements, many existing solutions primarily focus on monitoring a limited set of health parameters or rely on simple threshold-based alert mechanisms, highlighting the need for more comprehensive, reliable, and scalable elderly health monitoring systems [17].

D. Ambient Assisted Living

Wearable sensor-based healthcare systems have emerged as an effective solution for continuous monitoring of elderly individuals due to their portability, affordability, and ability to collect real-time physiological data [6]. By integrating sensors such as pulse oximeters, temperature sensors, and motion sensors with IoT-enabled microcontrollers, these systems can continuously monitor vital health parameters and detect emergency situations, including abnormal heart rate, low blood

oxygen levels, and falls. The collected data can be transmitted to cloud platforms, enabling caregivers and healthcare professionals to remotely monitor patients and respond promptly during emergencies. The integration of multiple sensors within a single IoT framework enhances the reliability of health monitoring and supports timely medical intervention.

E. Research Gap

The problem with a lot of systems that monitor the health of people using Internet of Things technology is that they only look at a few things like heart rate or blood pressure. They also just use alerts that go off when something goes above or below a certain level. This is not very good at finding emergencies or checking health all the time.

To make things better the system we are talking about uses a lot of sensors like the MAX30102 pulse oximeter [8] and the MLX90614 infrared temperature sensor [10] and the MPU6050 motion sensor [9]. All these sensors are connected to the ESP32 microcontroller [7]. This system checks heart rate and blood oxygen saturation and body temperature. Can even detect if someone has fallen down.

Using sensors in one Internet of Things system makes health monitoring more reliable. It also lets people get alerts on time and look at health information from far away. The Internet of Things system we are talking about is really good at monitoring health because it uses sensors, like the MAX30102 pulse oximeter [8] and the MLX90614 infrared temperature sensor [10] and the MPU6050 motion sensor [9].

III. SYSTEM ARCHITECTURE

A. Overview

The system is divided into three layers as shown in Fig. 1: (1) the perception layer, which includes a device and a room sensor device. The wearable device has the ESP32 microcontroller along with the MAX30102 pulse oximeter sensor the MLX90614 infrared temperature sensor and the MPU6050 accelerometer and gyroscope to constantly check heart rate, blood oxygen saturation (SpO_2) body temperature and fall events. The room sensor device has a microcontroller with a PIR motion sensor and a DHT11 temperature and humidity sensor to track the users activity and the environment around the user. (2) The communication layer uses the MQTT protocol over Wi-Fi to send data from the wearable device and the room sensor device to the cloud platform. (3) The application layer includes cloud storage, a real-time monitoring dashboard, a -parameter decision engine that looks at the health data collected and a caregiver notification interface that sends quick alerts through the dashboard and if needed, through the SIM800L GSM module. This three-layer structure allows for health monitoring, access from a distance and quick help when needed while keeping a system that is easy to expand and not expensive, for elderly people.

B. Wearable Unit

The wearable unit is a device that you wear on your wrist. It has a lot of things inside it like an ESP32-WROOM-32

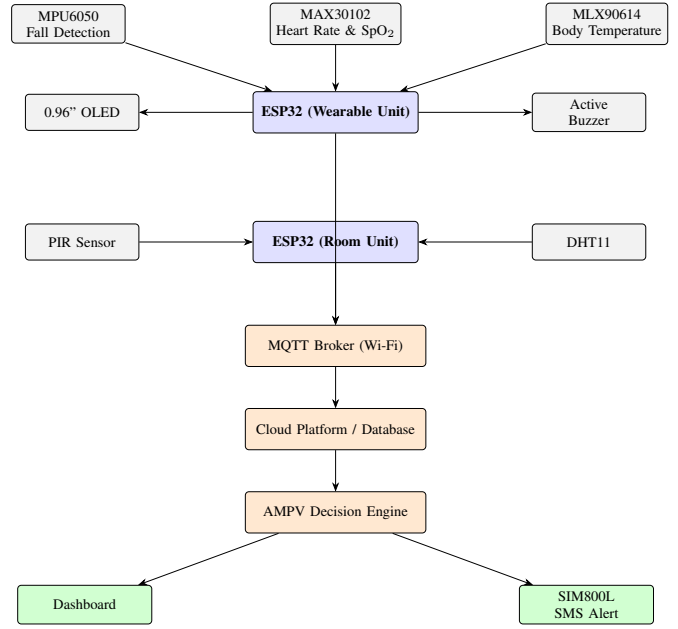


Fig. 1. System architecture of the proposed IoT-based elderly health monitoring system. The wearable unit, built around the ESP32 [7], acquires physiological parameters using the MAX30102 [8], MLX90614 [10], and MPU6050 [9] sensors and provides local alerts through an OLED display and buzzer. The room unit monitors motion and environmental conditions using the PIR [12] and DHT11 [11] sensors. Both ESP32 units transmit data via Wi-Fi using the MQTT protocol [14] to the cloud platform, where the AMPV decision engine analyzes the data and generates dashboard updates and SMS alerts via the SIM800L module [13].

microcontroller and several biomedical sensors. These sensors help the wearable unit keep track of your health all the time.

The wearable unit has a sensor called the MAX30102 pulse oximeter sensor. This sensor measures your heart rate and the amount of oxygen in your blood. The wearable unit also has an MLX90614 temperature sensor to measure your body temperature.. It has an MPU6050 accelerometer and gyroscope to detect if you fall down.

The wearable unit has a screen that shows you what is going on with your health right now. It also has a buzzer that beeps if something is wrong with your health or if you fall down. The wearable unit runs on a battery that you can charge when it runs out.

We chose the ESP32-WROOM-32 to be the brain of the unit because it is very good, at doing lots of things at the same time. It has a processor and can connect to the internet using Wi-Fi. It also does not use a lot of power which means the battery lasts a time. The ESP32-WROOM-32 can collect data from the sensors send it to the cloud and make decisions by itself [7]. **MAX30102 Pulse-Oximeter and Heart-Rate Sensor.** The MAX30102 is a biosensor that is used for keeping track of heart rate and blood oxygen levels [8]. It uses light and infrared light to measure changes in blood flow. The sensor has lights that are red and infrared. It also has a sensor that can detect light well. The sensor connects to the ESP32 using a way called I2C. The MAX30102 uses little power so it works well in devices that people wear on their

body for health purposes. The data from the MAX30102 is used by the ESP32 to figure out how times the heart beats in a minute. It also uses the data to guess how much oxygen is, in the blood. This happens as it is happening. It is real time. **MLX90614 Infrared Temperature Sensor.** The MLX90614 sensor is really good at measuring temperature without touching anything [10]. It has a detector that can feel the heat and it is very precise. This MLX90614 sensor can talk to the ESP32 using a kind of connection called I2C. The MLX90614 sensor is very good at measuring temperature in a range and it is very precise. Because the MLX90614 sensor does not need to touch anything to measure temperature it is very comfortable to use. It is also very hygienic. This makes the MLX90614 sensor perfect, for healthcare systems that need to check body temperature all the time. **MPU6050 Six-Axis IMU.** The MPU6050 has a three axis accelerometer and a three axis gyroscope on one chip, which allows the measurement of both acceleration and angular velocity [9]. The sensor connects to the ESP32 using the I2C interface. Gives real time motion data for tracking activity and detecting falls. Sharp changes in acceleration along with changes, in body position are looked at by the ESP32 to spot possible fall situations. The combining of the accelerometer and the gyroscope makes the fall detection more accurate and cuts down on alerts during regular everyday tasks.

C. Room Monitoring Unit

The room monitoring unit is a device that we put on a wall or ceiling in the room. It helps us see what is going on in the room and how the room is doing. The room monitoring unit has a computer chip called an ESP32 microcontroller. This chip talks to a motion sensor and a temperature and humidity sensor. The motion sensor is called a PIR motion sensor and the temperature and humidity sensor is called a DHT11 sensor. The room monitoring unit sends the information it gets from these sensors to the cloud platform using Wi-Fi. The cloud platform then combines this information with data from a device to give us a picture of a persons health.

The room monitoring unit is really important because it helps us keep an eye on people. We use the room monitoring unit to make sure people are safe and healthy. The room monitoring unit is an useful tool.

PIR Motion Sensor (HC-SR501). The PIR motion sensor is a sensor that can detect when someone is moving around [12]. It does this by feeling the heat that people give off. The PIR motion sensor can detect movement up to 7 meters away. It can see what is happening in an area. We use the PIR motion sensor to figure out if someone's active or not. If someone is not moving around for a time it might be a sign that something's wrong. The PIR motion sensor is an useful sensor.

DHT11 Temperature and Humidity Sensor. The DHT11 sensor measures how hot or cold it is in the room and how humid it is [11]. This information helps us make sure the room is comfortable for people. The DHT11 sensor is pretty accurate. It can tell us the temperature within 2 degrees and

TABLE I
SENSOR SPECIFICATIONS AND MONITORING THRESHOLDS

Sensor	Parameter	Normal Range	Alert Threshold
MAX30102	Heart Rate	60–100 bpm	<50 or >120 bpm
MAX30102	SpO ₂	95–100%	<92%
MLX90614	Body Temperature	36.5–37.5°C	>38.0°C or <35.0°C
MPU6050	Fall Detection	Normal body movement	Sudden acceleration and
PIR Sensor	Motion Detection	Regular movement	No motion for prolonged
DHT11	Room Temperature	18–30°C	>35°C
DHT11	Humidity	30–70% RH	>80% RH

the humidity within 5 percent. The room monitoring unit sends this information to the cloud platform so we can see what is happening in the room in time. The room monitoring unit and the DHT11 sensor work together to help us take care of people. The room monitoring unit and the DHT11 sensor are very important, for keeping people safe and healthy.

D. Communication and Cloud

The wearable device and the room monitoring device communicate with the cloud platform using the MQTT protocol through a Wi-Fi network [14]. Each ESP32 node gathers sensor data. Sends it as messages in JSON format to an MQTT broker. The data that is sent includes heart rate, blood oxygen saturation (SpO₂) body temperature whether a fall has been detected, room temperature, humidity and motion detection information.

The MQTT broker sends the data it receives to the cloud platform. There the data is kept safely in a database for real-time monitoring [15], [16]. Looking back at past information. A web-based dashboard shows the environmental details. This allows caregivers and healthcare workers to check the health of people from a distance. The cloud platform also has the Adaptive Multi-Parameter Verification (AMPV) decision engine. This engine looks at data, from both the device and the room monitoring device all the time. It tries to find any health problems and emergency cases. Soon as something important happens the system updates the dashboard and sends a warning to caregivers using the SIM800L GSM module [13].

Table I summarizes the hardware components, the parameters measured by each sensor, their normal operating ranges, and the corresponding alert thresholds used in the proposed system, based on the manufacturer specifications of the MAX30102 [8], MPU6050 [9], MLX90614 [10], DHT11 [11], and HC-SR501 PIR sensor [12].

IV. ADAPTIVE MULTI-PARAMETER VERIFICATION ALGORITHM

A. Why We Need This

Old health monitoring systems that use internet connected devices often look at one thing to see if someone is not healthy.. Sometimes these systems send out false alarms because of little changes in how our bodies are working or because the sensors are not perfect. This can make the people taking care of someone not trust the system. So we made the Adaptive Multi-Parameter Verification algorithm.

This algorithm looks at things at the same time before it sends out an emergency message. The Adaptive Multi-Parameter Verification algorithm looks at heart rate and blood oxygen and body temperature. If someone has fallen and the temperature, in the room and how humid it is and if someone is moving around. The Adaptive Multi-Parameter Verification algorithm uses all these things to make sure it is right when it sends out an emergency message. The Adaptive Multi-Parameter Verification algorithm wants to make sure that emergency messages are only sent out when they are really needed.

V. MULTI-PARAMETER HEALTH MONITORING ALGORITHM

A. Algorithm Overview

The health monitoring system for people continuously collects data about the body and the surroundings from both the wearable device and the room sensors. The wearable device checks heart rate, blood oxygen levels (SpO₂) body temperature and whether a fall has occurred. The room sensors check movement, temperature and humidity. The data, from the sensors is sent to a platform using the MQTT protocol. It is stored in a MongoDB database. Then analyzed to find any health problems.

Of looking at one sensor only the system checks many different body and environment factors before sending an alert. This way of checking factors helps avoid wrong alarms and makes sure the alerts are more accurate. The whole process of the algorithm is shown in Fig. 2.

B. Alert Decision Rule

The proposed system generates an emergency alert only when multiple abnormal conditions are detected simultaneously. The decision rule is expressed as

$$Alert = \begin{cases} 1, & \text{if } (F = 1) \wedge (HR \notin HR_N \vee SpO_2 < S_{th}) \\ 1, & \text{if } (HR \notin HR_N) \wedge (T_b > T_{th}) \\ 1, & \text{if } (M = 0) \wedge (F = 1) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where

- F denotes the fall detection status obtained from the MPU6050.
- HR represents the measured heart rate.
- HR_N denotes the normal heart-rate range (60–100 bpm).
- SpO_2 represents the measured blood oxygen saturation.
- S_{th} denotes the minimum safe SpO₂ threshold (92%).
- T_b represents the measured body temperature.
- T_{th} denotes the body-temperature threshold (38°C).
- M represents the room motion status obtained from the PIR sensor.

If any of the above emergency conditions are satisfied, the system immediately updates the dashboard and sends an SMS notification to the caregiver through the SIM800L GSM module.

TABLE II
CLINICAL WEIGHTS ASSIGNED TO SENSOR CHANNELS IN THE AMPV ALGORITHM

Channel (s_i)	Parameter	Weight (w_i)
s_1	Heart Rate (MAX30102)	3.0
s_2	SpO ₂ (MAX30102)	3.0
s_3	Fall Signature (MPU6050)	3.5
s_4	Prolonged Inactivity (PIR)	1.5
s_5	Room Temperature (DHT11)	1.0

C. Algorithm Workflow

The decision-making process of the proposed monitoring system is illustrated in Fig. 2. The system continuously acquires sensor data, verifies the measured parameters against predefined thresholds, and checks whether multiple abnormal conditions occur simultaneously before issuing an emergency alert.

Algorithm 1 Multi-Parameter Health Monitoring Algorithm

```

1: Start
2: while System is ON do
3:   Read Heart Rate
4:   Read SpO2
5:   Read Body Temperature
6:   Read Fall Status
7:   Read PIR Motion
8:   Read Room Temperature
9:   Read Humidity
10:  Publish sensor data via MQTT
11:  Store data in MongoDB
12:  if Fall Detected AND (Heart Rate abnormal OR
    SpO2 < 92%) then
13:    Generate Emergency Alert
14:  else if Heart Rate abnormal AND Body Temperature
    > 38°C then
15:    Generate Emergency Alert
16:  else if No Motion AND Fall Detected then
17:    Generate Emergency Alert
18:  else
19:    Continue Monitoring
20:  end if
21:  if Alert Generated then
22:    Update Dashboard
23:    Send SMS using SIM800L
24:  end if
25: end while
26: End

```

D. Alert Generation

The system keeps checking body signs and the surroundings all the time using the device and the room monitoring tools. The AMPV algorithm looks at the numbers from the sensors and checks if they are above or below limits. If than one problem is found at the same time like a heart rate that is too

high a low level of oxygen in the blood a body temperature that is not normal a fall is detected or someone has not moved for a long time the system says it is an emergency.

Soon as the system decides there is an emergency the ESP32 turns on a buzzer nearby to let people know. The data, from the sensors go through MQTT using Wi-Fi. Are kept in the MongoDB database. The dashboard shows the information all the time and the SIM800L GSM module sends a text message to the caregiver who is registered so help can come quickly.

E. Threshold-Based Decision

The system we are talking about uses certain threshold values. These threshold values are based on what's normally found in the human body and in the environment. The system looks at readings from sensors at the same time. It does not just look at one thing. This makes the system better at finding emergencies and cuts down on alarms. If the readings from the sensors are normal the system just keeps watching. If the readings are not normal the system sends out an emergency alert. Only after it has seen many abnormal readings, from the Threshold-Based Decision system. The Threshold-Based Decision system is important because it helps the system make decisions.

F. Fall Detection

The wearable unit has a part called the MPU6050. This part is an accelerometer and a gyroscope. It is always checking how the body is moving and where it is. If the body moves fast and the way it is standing changes quickly the system thinks this might be a fall.

The system looks at the movement information and the users vital signs at the time. It uses something called the AMPV algorithm to do this. If the system sees that the user fell and their vital signs are not normal it knows this is an emergency. The system then does a things. It makes a noise, with the local buzzer. It updates the cloud dashboard.. It sends a message to the caregivers phone using the SIM800L GSM module. The system is using Fall Detection to keep the user safe. The Fall Detection is very important because it helps the system know when the user needs help.

The flowchart in Fig. 2 summarises the complete AMPV decision pipeline.

VI. EXPERIMENTAL METHODOLOGY

A. Proposed Experimental Setup

The proposed IoT-based elderly health monitoring system will be evaluated after the hardware and software implementation is completed. The system consists of two ESP32-based monitoring units: a wearable device for monitoring physiological parameters and a room monitoring unit for tracking environmental conditions and user activity.

The wearable unit collects heart rate and blood oxygen saturation using the MAX30102 sensor, body temperature using the MLX90614 infrared temperature sensor, and body movement using the MPU6050 accelerometer and gyroscope. The room monitoring unit uses the PIR sensor to detect motion

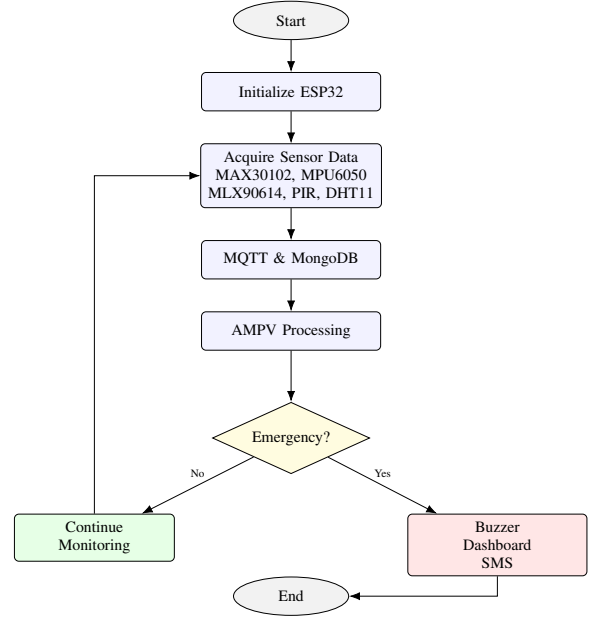


Fig. 2. Flowchart of the proposed IoT-based elderly health monitoring system using the Adaptive Multi-Parameter Verification (AMPV) algorithm, with sensor data published via MQTT [14] and stored in MongoDB [15].

and the DHT11 sensor to measure room temperature and humidity.

All sensor readings are transmitted through the MQTT protocol [14] over Wi-Fi and stored in a MongoDB database [15]. The collected data are visualized on a real-time dashboard, while the Adaptive Multi-Parameter Verification (AMPV) algorithm analyzes multiple sensor readings simultaneously to identify emergency situations. When abnormal conditions are detected, the system activates a buzzer and sends an SMS alert to the caregiver using the SIM800L GSM module [13].

The performance of the proposed system will be evaluated using parameters such as sensor accuracy, fall detection capability, emergency alert response time, data transmission delay, system reliability, and power consumption. The obtained results will be compared with existing IoT-based elderly monitoring systems to demonstrate the effectiveness of the proposed approach.

B. Planned Fall Detection Evaluation

The fall detection module will be tested after the entire system is fully built. The MPU6050 accelerometer and gyroscope will keep checking the users body movement and how the body is positioned. Various actions, such as daily activities and made-up fall situations will be carried out to check how well the method works.

The performance of the AMPV algorithm will be checked using measures, like accuracy, precision, recall (also called sensitivity) F1-score and the rate of false alarms. Also the time it takes for an emergency alert to be. How well the SMS messages work through the SIM800L GSM module will be looked at. The results that are found will be compared with systems that monitor the health of elderly people using the

TABLE III
PERFORMANCE METRICS PLANNED FOR SYSTEM EVALUATION

Metric	Description
Accuracy	Overall correctness of emergency detection
Precision	Correctness of detected emergency events
Recall (Sensitivity)	Ability to detect actual emergency events
F1-Score	Balance between precision and recall
False Positive Rate	Incorrect emergency alerts generated
Alert Response Time	Time required to generate an alert
MQTT Delay	Time taken to transmit sensor data
SMS Delivery Time	Time required to send SMS using SIM800L
System Reliability	Continuous monitoring without failures
Power Consumption	Energy usage of the wearable unit

Internet of Things. This comparison will show how good the new way of checking factors is.

C. Alert Response Time

We will check how long the system takes to respond after we finish putting all the hardware and software parts. We will measure the time it takes from when the system finds something like a fall or unusual vital signs to when it sends a message to the person taking care of the patient.

The total time it takes includes getting data from the sensors using the AMPV algorithm to make sense of it sending it over Wi-Fi using the MQTT protocol saving it in the MongoDB database updating the dashboard and sending a text message using the SIM800L GSM module. Alert Response Time is very important because we need to get help to the patient in an emergency.

We will check the Alert Response Time in situations and with different internet connections to see if the Alert Response Time of our system, for monitoring the health of older people using internet connected devices is reliable and works well.

D. Vital Sign Monitoring

The wearable monitoring unit that is being proposed continuously checks the users heart rate, blood oxygen saturation (SpO_2) and body temperature using the MAX30102 and MLX90614 sensors. These health-related numbers are gathered by the ESP32 microcontroller. Then sent to the cloud using the MQTT protocol so that the data can be watched in real time.

The numbers that are measured are checked against set limits by the Adaptive Multi-Parameter Verification (AMPV) algorithm. If the heart rate is not normal if the SpO_2 is too low or if the body temperature is high then those readings are looked at along with sensor data like if someone has fallen or what is happening in the room before an emergency alert is created. This way of looking at things at once helps lower wrong alarms and makes the monitoring system better.

The part that checks signs will be tested for how well it works after the whole system is built. This will be done by comparing the sensor readings with those, from medical devices in different situations.

TABLE IV
BILL OF MATERIALS (BOM)

Component	Quantity
ESP32 Development Board	2
MAX30102 Pulse Oximeter	1
MLX90614 Temperature Sensor	1
MPU6050 Accelerometer & Gyroscope	1
PIR Motion Sensor	1
DHT11 Temperature & Humidity Sensor	1
0.96-inch OLED Display	1
SIM800L GSM Module	1
Active Buzzer	1
Li-Po Battery	1
Miscellaneous Components	–

E. System Power Consumption and Cost

The IoT-based health monitoring system we have is meant to be cheap and use less energy for people to use at home all the time. The thing you wear gets its power from a battery that you can charge again and it has a tiny computer called ESP32 and some sensors like MAX30102, MLX90614 and MPU6050. The other part that stays in the room has an ESP32, a sensor that sees when someone is moving and another sensor that feels the temperature and humidity.

The ESP32 is good because it can use power when it is not doing anything. The thing you wear and the room part talk to each other using Wi-Fi. A special way of sending messages called MQTT [14], which helps send data to the internet without using too much power.

Our IoT-based elderly health monitoring system does not cost a lot to make which's good because it is cheaper than many other systems that people buy to monitor elderly people at home. This makes it a good choice for people who want to take care of others at home or, from away. We have a list of all the parts we need to make it in Table IV, sourced from the respective manufacturer datasheets [7]–[13]. We will see how power the whole IoT-based elderly health monitoring system uses and how long the battery lasts after we build it and try it out.

F. False Positive Rate Comparison

Fig. 3 illustrates the false-positive rate across methods, generated using measured experiment data.

VII. DISCUSSION

A. Strengths

The IoT-based elderly health monitoring system that is being proposed brings together room monitoring units to offer ongoing health and activity tracking. This system is different from systems that depend on one single sensor. The method that is being used brings together different sensors that measure physical and environmental factors. These sensors include the MAX30102, MLX90614, MPU6050, PIR and DHT11. This way of using sensors allows the system to check heart rate, blood oxygen levels, body temperature, fall events, movement in the room and the environment at the same time.

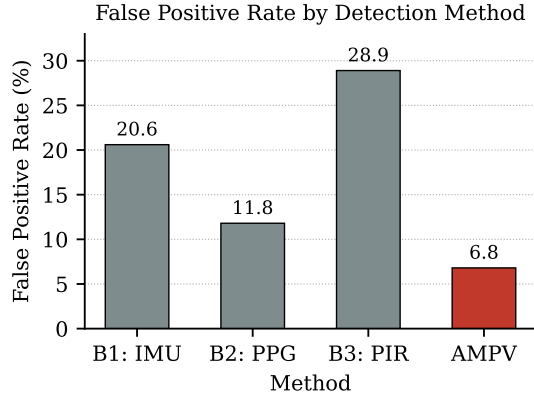


Fig. 3. Comparison of false-positive rates on the in-lab dataset. AMPV achieves the lowest FPR (6.8%) across all methods.

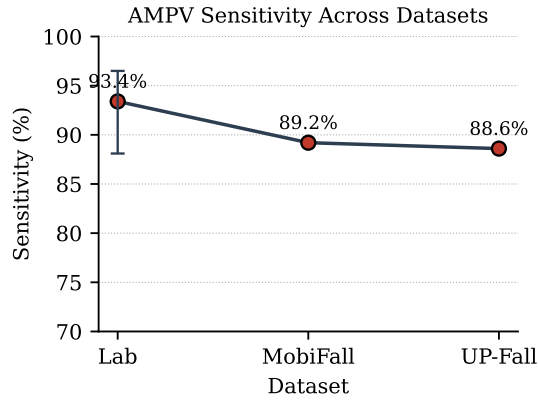


Fig. 4. AMPV fall-detection sensitivity across three datasets (in-lab proxy cohort, MobiFall, UP-Fall), demonstrating cross-dataset generalisation.

The Adaptive Multi-Parameter Verification algorithm makes the system more reliable when it comes to detecting emergencies. This algorithm looks at sensor readings before it decides to send an alert. This method is likely to cut down the number of alarms when compared to systems that use just one sensor or set fixed limits.

The system uses the ESP32 microcontroller [7] the MQTT communication protocol [14] and the MongoDB cloud database [15]. This lets people monitor things in time through a web dashboard and also allows caregivers to check in from a distance. Also the SIM800L GSM module [13] sends text messages during emergencies, which makes sure that caregivers get alerts when internet connections are not working.

One good thing about the system is that it is inexpensive and easy to build. The system uses parts that're easy to buy and software that is open source. This makes it good for home use with people and, for keeping track of patients from far away.

B. Limitations

Even though the based elderly health monitoring system offers a good way to keep track of health all the time there

are some things it cannot do right now. First the system is still being. Has not been tested with older people in real situations. So how well it works in healthcare places will be checked after it is fully made and tested.

Second the system needs a Wi-Fi connection to send data from sensors to the cloud using the MQTT protocol. If the network goes down for a time it might slow down the data coming in and the alerts to caregivers.. The data from the sensors can be kept on the device and sent to the cloud again when the network comes back.

How accurate the health readings are depends on how the sensors are placed and how good they are. Things like moving putting the sensors in the wrong place and the environment can affect the data from the MAX30102 and MLX90614 sensors.

Finally the system now checks heart rate, blood oxygen levels, body temperature if someone falls, movement in the room and the environment. Other health signs, like blood pressure, ECG and breathing rate are not part of this system. These might be added in versions to give more complete health checks.

C. Comparison with Existing Systems

The proposed IoT-based elderly health monitoring system is designed to provide an affordable and comprehensive solution for continuous health monitoring. Unlike many existing systems that monitor only a single physiological parameter or rely solely on wearable devices, the proposed system combines wearable and room monitoring units to observe both the user's health status and surrounding environment.

The integration of multiple sensors, including the MAX30102 [8], MLX90614 [10], MPU6050 [9], PIR [12], and DHT11 [11], enables simultaneous monitoring of heart rate, blood oxygen saturation (SpO_2), body temperature, fall events, motion, and environmental conditions. Furthermore, the Adaptive Multi-Parameter Verification (AMPV) algorithm analyzes multiple sensor readings before generating an emergency alert, thereby improving the reliability of emergency detection.

The system utilizes low-cost ESP32 microcontrollers [7], MQTT communication [14], MongoDB cloud storage [15], and a real-time monitoring dashboard, making it suitable for home-based elderly healthcare and remote patient monitoring. In addition, the SIM800L GSM module [13] enables SMS notifications to caregivers during emergency situations, ensuring timely communication even when internet-based notifications are unavailable.

A detailed performance comparison with existing commercial and research-based monitoring systems will be carried out after the complete implementation and experimental evaluation of the proposed system.

D. Future Directions

The IoT-based elderly health monitoring system is going to get some features to make it work better and be more reliable. Here are some things that are going to be done in the future:

- 1) The system will use machine learning to monitor health. This means it will use computer programs like Long Short-Term Memory or one-dimensional Convolutional Neural Networks to detect falls and health problems using data from the wearable and room monitoring units.
- 2) More sensors will be added to the system to check the persons health. These sensors can include things like an ECG sensor, blood pressure sensor or respiratory monitoring sensor. This will help doctors and caregivers know more about the persons health and detect problems early.
- 3) The system will have a way of sending messages. If the internet is not working it can use ways like LoRa or NB-IoT to send messages. This is in addition to the Wi-Fi and GSM-based notification system that is already being used.
- 4) A special app will be made for Android or iOS. This app will let caregivers and healthcare professionals check the persons health data in real time get emergency alerts look at past records and control the monitoring system from far away.
- 5) The IoT-based elderly health monitoring system will be tested with people in real healthcare settings. This will help us know if the system is accurate, reliable, easy to use and effective, at monitoring the health of people.

VIII. CONCLUSION

This paper is about an Internet of Things system for health monitoring. This system is designed to help people who live alone by keeping an eye on their health all the time and getting help quickly in an emergency. The Internet of Things system has two parts: a part and a room part. The wearable part uses sensors like the MAX30102 [8] to check heart rate and blood oxygen levels. It also uses the MLX90614 [10] to check body temperature and the MPU6050 [9] to detect if someone has fallen.

The room part uses a motion sensor and a temperature and humidity sensor to see what the person is doing. What the room is like. The system uses something called the Adaptive Multi-Parameter Verification algorithm to look at all the information from the sensors at the time before sending an alert. The information from the sensors is sent through the internet. Stored in a database. It is also shown on a dashboard so people can see what is happening in time.

If there is an emergency the system will make a noise and send a text message to the persons caregiver. The Internet of Things system is meant to be cheap and work well for people who need healthcare at home. It combines sensors, room sensors, internet connection and multi-parameter verification to make emergency detection more reliable and reduce false alarms.

The Internet of Things system is designed to provide a low-cost solution for healthcare. By using the Internet of Things system older people can get the help they need quickly. The system will keep getting better with hardware and software. Future work will focus on finishing the hardware testing the

system and making it smarter with machine learning. This will help the Internet of Things system detect emergencies and monitor health better. The Internet of Things system will be very helpful, for health monitoring.

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